Quantum-Assisted Hypothesis Search

Research-readiness extension | 64 hypotheses + compact QUBO + matched conditions

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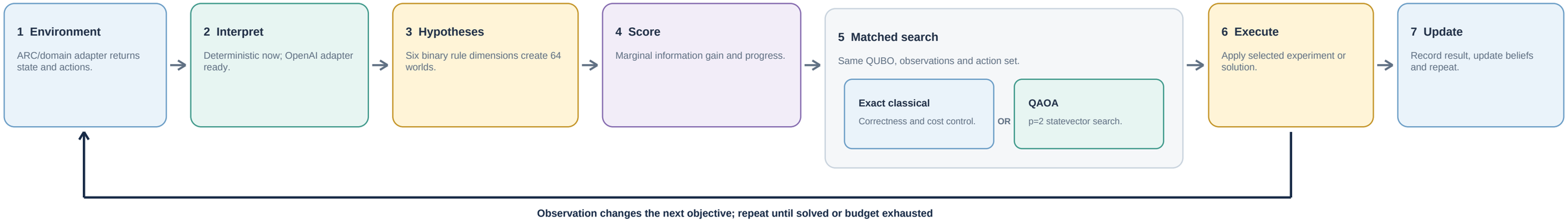
RUNTIME RESEARCH LOOP



Observation changes the next objective; repeat until solved or budget exhausted

WHAT IS IMPLEMENTED

Factorized world model

factorized.py + rule_environment.py

Player, Goal, Movement, Topology, Interaction and State. Two options each produce 2^6 = 64 complete world models.

Compact optimization model

factorized_objective.py + qubo.py

Information gain comes from six marginals. QUBO uses six hypothesis bits and seven one-hot action variables.

Matched experiment runner

factorized_agent.py + benchmark.py

Policy proxy, exact classical search and QAOA receive identical states, utilities and action choices.

Provider and environment slots

providers/ + environment_adapter.py

OpenAI Responses API adapter reads environment variables. External ARC or domain objects can be wrapped later.

PRELIMINARY ENGINEERING RUNS

Condition	Worlds	Solve rate	Mean actions	Mean wall time
Policy proxy	64	100%	7.000	~0.007 s
Exact classical	64	100%	7.000	~0.007 s
QAOA smoke	8	100%	6.875	~3.26 s

Smoke tests validate the implementation only. The QAOA configuration was developed on small samples; no advantage claim is justified.

SCALING ESTIMATE FOR CURRENT ONE-HOT ACTION FORMULATION

Hypotheses	H bits	Total qubits	Statevector	Raw memory
64	6	13	8,192	128 KiB
256	8	15	32,768	512 KiB
262,144	18	25	33.6M	512 MiB
16,777,216	24	31	2.15B	32 GiB

Memory excludes simulator overhead, ancillas and transpilation. Real feasibility may fail before the raw qubit count.

CURRENT BOUNDARY

The QUBO is compact, but statevector simulation still expands its complete cost diagonal. This is a correctness prototype, not evidence of quantum speedup.

Raw outputs: output/experiments/ | Method notes: EXPERIMENT_NOTES.md | API key is never stored in source or traces.

Quantum-Assisted Hypothesis Search

Complete file map and responsibilities

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RESEARCH PATH

64-HYPOTHESIS MODEL

src/qahs/factorized.py
Six rule dimensions, actions, hypotheses, states and traces.
src/qahs/rule_environment.py
Controlled hidden-world environment for matched experiments.
src/qahs/environment_adapter.py
Contract and wrapper for future ARC or domain environments.
src/qahs/factorized_objective.py
Marginal information gain, action scores and belief updates.
src/qahs/qubo.py
QUBO model, Ising conversion, constraints and resource counts.
src/qahs/factorized_agent.py
Observe-search-act-update loop for the 64-world study.

EXPERIMENTS + SCALE

src/qahs/benchmark.py
Repeated matched runs and aggregate metrics.
src/qahs/benchmark_cli.py
Command-line benchmark entry point.
src/qahs/scaling.py
Qubit, amplitude and statevector-memory estimates.
output/experiments/*.json
Preserved row-level results and summaries.
output/experiments/README.md
Plain-language index of result files.
EXPERIMENT_NOTES.md
Interpretation, tuning history and remaining requirements.

SEARCH + PROVIDERS

src/qahs/search/base.py
Shared search backend contract.
src/qahs/search/greedy.py
Immediate-utility policy proxy.
src/qahs/search/classical.py
Exact minimum over valid encoded states.
src/qahs/search/qaoa.py
Cost phase, X mixer, COBYLA and sampled decoding.
src/qahs/providers/openai_provider.py
Optional Responses API caller using environment credentials.
.env.example
Variable names only; never contains a real key.

INTERFACES + VERIFICATION

dashboard/server.py
Dependency-free local run dashboard and JSON endpoint.
dashboard/app.py
Optional Streamlit run/replay interface.
tests/test_qahs.py
Nine checks for encoding, objectives, QUBO/Ising and loops.
pyproject.toml
Package, scripts and optional dependency groups.
README.md
Setup, commands, boundaries and project map.
.gitignore
Excludes credentials, caches and generated traces.

FOUNDATIONAL THREE-QUBIT DEMO

DOMAIN + LOOP

src/qahs/models.py
Original actions, states, hypotheses, scores and trace records.
src/qahs/environment.py
Red-tile switch demonstration.
src/qahs/interpreter.py
Deterministic and structured-LLM interpreter interfaces.
src/qahs/compiler.py
Binary encode/decode for hypothesis-action pairs.
src/qahs/agent.py
Original two-turn harness.

PREDICTION + STORAGE

src/qahs/predictor.py
Predicted observations and progress values.
src/qahs/objective.py
Entropy, information gain and posterior updates.
src/qahs/run_store.py
Replayable JSON trace storage.
src/qahs/cli.py
Three-qubit command-line runner.
src/qahs/__init__.py
Public package exports.

ARTIFACTS + SOURCES

output/pdf/
Current codebase overview.
tmp/pdfs/
Reproducible PDF build source.
Instructor QAOA notebook
Source pattern for phase, mixer, COBYLA and sampling.
Not reused
Malware graph, similarity weights and Max-Cut objective.
Dashboard URL
http://127.0.0.1:8000

RUN CONTROLS

qahs-demo: 3-qubit loop | qahs-benchmark: 64-hypothesis matched study | dashboard/server.py: local browser interface

Next gate: real LLM + genuine ARC tasks + pre-registered held-out evaluation. Current runs cannot establish stronger AI or quantum advantage.